

Electric Storage in the Energy Transition

Projection of the Levelized Cost of Energy Storage, 2020-2030

Undergraduate thesis, English portfolio edition

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Translation Note

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Abstract

Electric storage could accelerate the energy transition; however, its future competitiveness remains highly uncertain. This research analyzes the lithium-ion battery value chain and the main public policies that, at the global level, are aimed at utility-scale lithium-ion batteries.

The thesis also proposes a methodology for calculating and projecting the Levelized Cost of Energy Storage (LCOS) as a function of hourly discharge duration. This approach seeks to estimate economies of scale more accurately than methods that assume a fixed discharge capacity.

The results show a 46 percent reduction in LCOS between 2020 and 2030, reaching approximately USD 63/MWh for a four-hour battery. This would imply a more competitive battery for long-duration services in isolated markets and in systems with high penetration of renewable energy.

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1. Introduction

At the end of the 2010s, expectations about the potential role of electric storage in the energy transition were gaining strength. In December 2019, the Nobel Prize in Chemistry recognized the main developers of the lithium-ion battery, a technology described as capable of storing significant amounts of solar and wind energy and making a fossil-fuel-free society possible.

The rapid growth of lithium batteries as stationary electric-storage systems has produced a large body of industrial and institutional reports evaluating their competitiveness across electricity-market applications and services. These studies have been complemented by research on utility-scale lithium battery adoption at global, regional, and national levels.

A central concept in this discussion is the Levelized Cost of Energy Storage, or LCOS: the average cost of storing one unit of electricity. The LCOS methodology is not yet fully standardized, and different approaches vary according to the technology and market service being analyzed.

Prior literature had offered empirical interpretations of the factors influencing cost reductions in lithium-ion batteries. However, at the time the thesis was written, no study known to the author had projected the LCOS of a utility-scale lithium battery using a methodology designed to capture economies of scale as discharge duration increases.

The thesis is therefore organized around two objectives: first, to describe the market, technological, supply-chain, and policy conditions shaping lithium-ion batteries; and second, to project battery-pack prices and LCOS from 2020 to 2030 under a framework that accounts for discharge capacity.

2. The Lithium Battery in the Electricity Market

Decarbonization is one of the most urgent long-term issues on the global agenda. The electricity sector is especially relevant because it is one of the largest sources of CO₂ emissions and because future electrification of mobility is expected to increase global electricity demand.

The expansion of variable renewable energy, particularly solar and wind, changes how electricity systems operate. Since these technologies have very low marginal costs, they enter the dispatch order before thermal plants. This can reduce market prices through the merit-order effect, but it also creates technical challenges when generation varies rapidly during the day.

One of the most visible examples is the so-called duck curve, observed in systems with high solar penetration such as California. During midday, solar generation reduces net load, while in the evening the system must quickly ramp up firm generation as solar output declines and demand remains high.

These dynamics create reliability and efficiency problems. Thermal generation may need to operate away from its most efficient range, while system operators must maintain frequency and balance supply and demand instantaneously. In this context, electric storage can provide flexibility to the grid.

Lithium-ion batteries can charge when electricity is abundant or inexpensive and discharge when it is scarce or expensive. This enables energy arbitrage, peak shaving, frequency regulation, reserve services, and support for higher renewable penetration.

In isolated or constrained systems, such as Baja California Sur in Mexico, these characteristics are especially important. High and volatile marginal prices can make storage more attractive than in larger interconnected systems, particularly where solar resources are abundant.

2.1 Background

Utility-scale electric storage includes several technologies: pumped hydro, compressed air, flywheels, flow batteries, lead-acid batteries, and lithium-ion batteries. Historically, pumped hydro dominated installed capacity, but lithium-ion batteries became the fastest-growing technology during the 2010s.

The United States became a leading market for lithium battery deployment. From 2011 to 2019, lithium batteries were the technology that added the most new utility-scale storage capacity to the U.S. grid, with installed capacity growing many times over in less than a decade.

Project sizes also increased globally. Important battery projects emerged in South Korea, Australia, Japan, China, and California, showing that lithium-ion technology was moving from demonstration scale toward larger utility-scale applications.

Although other storage technologies reduced costs during the decade, lithium-ion stood out for sustained cost declines and technical improvements. This positioned it strongly against other emerging storage technologies for the 2020s.

2.2 Value Chain

The thesis analyzes the utility-scale lithium battery value chain through four main stages: extraction, manufacturing, commercialization, and recycling.

Extraction is often identified as the greatest short-term risk in the lithium-ion battery value chain. Concerns focus on the availability and geographic concentration of minerals such as lithium and cobalt. In 2019, cobalt production was highly concentrated in the Democratic Republic of Congo, while lithium reserves were concentrated in countries such as Argentina, Australia, and Chile.

Mineral-price volatility can affect the industry. For example, expectations of higher battery demand and uncertainty around cobalt availability contributed to a large increase in cobalt futures prices between 2017 and 2018. Still, studies cited in the thesis suggest that even large increases in individual mineral prices translate into relatively smaller increases in total battery-pack costs.

Manufacturing requires the integration of cells, modules, battery packs, inverters, containers, cooling systems, software, and battery management systems. Battery-cell chemistry matters: NMC, NCA, and LFP cathode technologies differ in cost, energy density, safety, and exposure to critical minerals.

Commercialization depends on regulation, market design, and the ability of batteries to monetize services. Storage can provide value through capacity, ancillary services, energy arbitrage, congestion relief, renewable integration, and reliability support, but revenues depend on whether markets recognize these services.

Recycling and second-life applications become increasingly important in the long run. As electric-vehicle batteries reach the end of their first life, they may be reused in stationary applications or recycled to recover valuable materials. Public policy in the European Union and the United States has begun to support recycling capacity and innovation.

2.3 Public Policy

Public policy has been central to the development of lithium-ion batteries and energy storage. Governments have supported the industry through research and development, manufacturing incentives, electric-vehicle adoption programs, renewable-energy targets, and market rules for storage participation.

In the United States, federal and state actions have been especially important. FERC orders and state-level procurement targets have helped open electricity markets to storage resources. California, New York, Massachusetts, New Jersey, Nevada, Oregon, and Virginia are examples of jurisdictions that introduced targets or programs to accelerate deployment.

Internationally, several governments have used direct support, innovation programs, and industrial strategies to encourage energy-storage adoption. These policies matter because battery costs decline through cumulative production, manufacturing scale, learning-by-doing, and technology improvement.

The thesis concludes that public policy can accelerate electric-storage adoption when it recognizes the unique attributes of storage rather than treating it only as generation, load, or transmission.

3. Literature Review

The literature review focuses on two analytical tools: experience curves and the Levelized Cost of Energy Storage.

Experience-curve theory is grounded in the idea that technological change is partly driven by learning. Wright's early work showed that production costs can decline as cumulative output increases. In energy technologies, this relationship is commonly estimated with a log-log model linking cost to cumulative production.

For lithium-ion batteries, experience curves are useful because battery-pack prices fell dramatically during the 2010s while cumulative production increased due to electric vehicles and stationary storage. The thesis applies this logic to forecast future battery-pack prices.

LCOS, in turn, provides an average cost of storing electricity over the lifetime of a storage asset. It is analogous to the Levelized Cost of Energy, but adapted to storage. A robust LCOS estimate must consider capital costs, fixed and variable operation and maintenance, replacement costs, efficiency, degradation, discount rates, cycles, and useful life.

A key limitation of some LCOS approaches is that they assume a fixed discharge duration. The thesis emphasizes that changing discharge duration affects cost components differently, so a methodology that captures discharge-duration economies of scale is needed for utility-scale batteries.

4. Methodology

The methodology has two main steps. First, the thesis forecasts the battery-pack energy component using an experience-curve model. Second, it uses the projected battery-pack cost as an input to estimate LCOS across different discharge durations from 2020 to 2030.

The experience-curve model uses cumulative battery-pack production as the explanatory variable and historical battery-pack prices as the dependent variable. Projected cumulative production is informed by electric-vehicle adoption scenarios, particularly the International Energy Agency's 30@30 scenario.

The LCOS model divides the real cost of the asset by the net energy that the battery can deliver over its useful life. Unlike a fixed-duration approach, the thesis adjusts the energy and power components according to discharge duration so that economies of scale can be observed as the battery is configured for longer service.

The methodology therefore connects a macro view of global battery manufacturing with a project-level measure of storage competitiveness.

4.1 Results

The battery-pack forecast is optimistic relative to some early observations because it is linked to a scenario in which electric mobility expands consistently with sustainable-development objectives. The thesis argues that the combination of value-chain advances, public policy, electric-vehicle incentives, and research-and-development investment makes a battery-pack price below USD 100/kWh by 2030 highly probable.

The LCOS forecast shows that cost reductions are not uniform across battery configurations. Longer discharge durations benefit from economies of scale because not all components rise proportionally with energy capacity.

The central result is a 46 percent LCOS reduction between 2020 and 2030. For a four-hour lithium-ion battery, the projected LCOS reaches approximately USD 63/MWh by 2030.

This suggests that utility-scale lithium-ion batteries may become increasingly competitive for longer-duration services, particularly in isolated systems or systems with high variable-renewable penetration.

5. Discussion and Conclusion

The thesis argues that lithium-ion batteries can play a significant role in the energy transition by providing flexibility, improving renewable integration, and reducing the reliability and efficiency problems created by high renewable penetration.

Storage is not valuable only because it stores electricity. Its economic value depends on where it is located, which services it can provide, what market rules allow it to monetize, and how quickly costs continue to decline.

Markets with high price volatility, grid constraints, isolated systems, or high solar penetration may see storage become competitive earlier. Baja California Sur is highlighted as an example where high marginal prices and abundant solar resources make storage especially relevant.

The research concludes that public policy and market design should recognize the unique attributes of electric storage. A regulatory framework that properly values storage services could accelerate clean-energy adoption while producing economic benefits for generators, consumers, and the electricity system.

By 2030, sustained cost declines, larger manufacturing scale, electric-vehicle growth, and supportive public policy could make utility-scale lithium-ion batteries a central resource for electricity systems with increasing renewable generation.

Selected Abbreviations

BESS: Battery Energy Storage System

BNEF: Bloomberg New Energy Finance

CAISO: California Independent System Operator

DOE: United States Department of Energy

EIA: Energy Information Administration

ESS: Energy Storage System

EV: Electric Vehicle

GHG: Greenhouse Gases

IEA: International Energy Agency

IPCC: Intergovernmental Panel on Climate Change

IRENA: International Renewable Energy Agency

LCOS: Levelized Cost of Energy Storage

MEM: Wholesale Electricity Market

PV: Photovoltaic

VRE: Variable Renewable Energy

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